

Causal Discovery under Non-Gaussianity: Robustness of LiNGAM under Violations of Independence

Simone Alberto Distefano

Carlos Rubiano

Abstract

We study the robustness of the DirectLiNGAM algorithm (Shimizu et al., 2011) under systematic violations of its key assumptions, using a comprehensive suite of controlled Monte Carlo experiments. While non-Gaussianity is generally sufficient for causal identification when disturbances are mutually independent, we show that dependence among disturbances can cause DirectLiNGAM to fail systematically—recovering the *wrong* causal direction even as sample size grows. We identify two qualitatively different failure modes. First, a common latent volatility factor induces dependence whose effect depends critically on the distributional shape of the idiosyncratic shocks: elliptical (Gaussian) and light-tailed symmetric (uniform) shocks lead to failure, while heavy-tailed or skewed shocks allow identification to be recovered. We further demonstrate that this failure is caused by the *unobserved* nature of the common factor: conditioning on the volatility factor fully restores identification. Second, when the error does not satisfy mutual independence condition but rather a weaker assumption such as mean independence—a condition of the form $\mathbb{E}[\varepsilon_Y | X] = 0$ but $\varepsilon_Y \not\perp X$ —DirectLiNGAM fails regardless of sample size and the strength of the violation, pointing to a structural limitation of the independence-based identification strategy. These findings connect directly to the recent theoretical literature showing that independence-based methods can fail at the population level (Mesters et al., 2024), and motivate the use of weaker mean-independence conditions as in the LiMIAM framework.

Contents

1	Introduction	2
2	Model and Assumptions	3
2.1	LiNGAM assumptions	3
2.2	Non-Gaussianity and ellipticity	3
2.3	Mean independence	3
3	Simulation Design	4
3.1	Scenario A: Gaussian baseline	4
3.2	Scenario B: Non-Gaussian independent disturbances	4
3.3	Scenario C: Common volatility	5
3.4	Scenario D: Mean independence but not independence	5

4	Results	6
4.1	Scenario A: Gaussian baseline	6
4.2	Scenario B: Non-Gaussian independent disturbances	6
4.3	Scenario C: Common volatility	7
4.4	Scenario D: Mean independence	9
4.4.1	Effect of dependence strength γ	9
5	Connection to Recent Theoretical Work	11
6	Conclusion	11

1 Introduction

Recovering the causal direction from observational data is one of the central problems in statistics and econometrics. In linear Gaussian structural models, causal link is fundamentally non-identifiable: both $X \rightarrow Y$ and $Y \rightarrow X$ generate the same joint distribution, leaving no statistical basis for choosing between them.

The Linear Non-Gaussian Acyclic Model (LiNGAM) of [Shimizu et al. \(2006\)](#) resolves this issue by exploiting the asymmetry that non-Gaussianity and independence of disturbances introduce. Under linearity, acyclicity, and the assumption that structural errors are mutually independent and non-Gaussian, the causal DAG is identifiable from the observational distribution. The approach connects causal discovery with independent component analysis (ICA) ([Comon, 1994](#)) and has produced an extensive literature, including the direct residual based variant DirectLiNGAM ([Shimizu et al., 2011](#)), pairwise-likelihood extensions ([Hyvärinen and Smith, 2013](#)) and applications to structural vector autoregressions ([Lanne et al., 2017](#)).

Despite the theoretical and empirical success of these methods, one key assumption is, in practice, fragile and often impossible to justify. The mutual independence assumption underlying the LiNGAM methodologies is a strong requirement on all conditional moments of the process, and it is difficult to defend in many settings. In macroeconomics and financial economics, structural shocks routinely exhibit conditional heteroskedasticity and volatility clustering ([Engle, 1982](#); [Bollerslev, 1986](#)). For example, one could argue that shocks to interest rates and shocks to inflation are each individually mean-unpredictable from past (or, trivially, future) values, yet their conditional variances co-move strongly over time. During episodes of heightened macroeconomic uncertainty—such as the 2008 financial crisis or the post-pandemic tightening cycle—both shocks tend to increase simultaneously in magnitude, driven by a common latent volatility factor, such as aggregate financial stress. Bearing this in mind, a commonly used and less restrictive assumption among econometricians is that $(\mathbb{E}[\varepsilon_i | \varepsilon_j] = 0)$ even while failing to be mutually independent.

[Mesters et al. \(2024\)](#) show theoretically that standard LiNGAM implementation, including DirectLiNGAM, among others ([Cardoso and Souluoumiac, 1996](#); [Hyvärinen, 1999](#); [Shimizu et al., 2011](#))—can recover the wrong causal direction at the population level when disturbances are mean-independent but not fully independent. They propose the Linear Mean-Independent Acyclic Model (LiMIAM) and the DirectLiMIAM algorithm as robust alternatives.

The goal of this project is to provide a systematic empirical characterization of these failure modes. Through controlled simulations, we identify the key distributional and dependence conditions under which DirectLiNGAM success and fails, quantify how performance degrades as assumptions are relaxed, and illustrate the conceptual distinctions that the theoretical literature highlights.

Our experiments cover four scenarios — a Gaussian benchmark, ideal non-gaussian plus independent shocks, common volatility, and mean independence— spanning a range of distributional shapes and dependence structures.

The project is organized as follows. Sections 2 presents the model and assumptions. the section 3 describes the simulation design. Section 4 reports and interpret the results. Section 5 connects the findings to recent work. Section 6 concludes.

2 Model and Assumptions

We consider the standard bivariate linear structural equation model (SEM):

$$X = \varepsilon_X, \quad Y = \beta X + \varepsilon_Y \quad (1)$$

with $\beta = 0.8$ throughout. The true causal directions is $X \rightarrow Y$ by construction. We define success as the event that DirectLiNGAM correctly identifies this direction.

2.1 LiNGAM assumptions

The LiNGAM model of Shimizu et al. (2006) rests on four assumptions:

1. *Linearity* The structural equations are linear in the parameters. $\mathbb{E}[Y | X] = \beta X$
2. *Acyclic structure* The induced graph is a directed acyclic graph (DAG).
3. *Independent disturbances* The errors ε_X and ε_Y are mutually independent.
4. *Non-Gaussian disturbances* At most one component of $(\varepsilon_X, \varepsilon_Y)$ is Gaussian.

Under this conditions, the causal order is identifiable and can be recovered by ICA.

2.2 Non-Gaussianity and ellipticity

Non-Gaussianity alone is not sufficient for identification. The relevant requirement is that the joint distribution of $(\varepsilon_X, \varepsilon_Y)$ is non-*elliptical*. Elliptical distributions— including the Gaussian, Student-t and any scale mixture of a spherical distribution— are invariant under any orthogonal rotation after whitening, so they do not provide enough asymmetry to distinguish causal direction. In contrast non-elliptical distributions (e.g. Laplace, exponential) break this symmetry, allowing identification.

As a concrete example, the common-volatility model $\varepsilon_i = \sigma U_i$ where U_1, U_2 are i.i.d. Gaussian produces an elliptical joint distribution regardless of the distribution of σ . this explains why Gaussian mixtures sit precisely at the boundary of the identifiability condition in Mesters et al. (2024) (Theorem 2.1 therein).

2.3 Mean independence

A weaker exogeneity condition that replaces full independence is *mean independence*:

$$\mathbb{E}[\varepsilon_Y | X] = 0. \quad (2)$$

This requires only that the conditional mean of ε_Y given X be zero, while allowing the full conditional distribution of ε_Y to depend on X through higher moments. For example, in the model

$\varepsilon_Y = \gamma XZ$ where $Z \perp X$ and $\mathbb{E}[Z] = 0$, condition (2) holds but $\text{Var}(\varepsilon_Y | X) = \gamma^2 X^2 \text{Var}(Z)$ depends on X , so $\varepsilon_Y \not\perp X$.

DirectLiNGAM requires full independence (2) strengthened to stochastic independence. The purpose of our simulation study is to document the practical consequences of relying on this stronger assumption.

3 Simulation Design

All experiments use the structural model (1) with $\beta = 0.8$, sample sizes $n \in \{100, 300, 1000\}$, and $M = 300$ Monte Carlo replications per cell. the success rate is the fraction of replications in which DirectLiNGAM (with pairwise likelihood score of Hyvärinen and Smith 2013) recovers the true directions $X \rightarrow Y$. We implement DirectLiNGAM via the `lingam` Python package (Ikeuchi et al., 2023)

We design four groups of experiments, each isolating a specific mechanism.

3.1 Scenario A: Gaussian baseline

We start from this benchmark to verify that our implementation correctly reproduces the known non-identifiability result. In this first scenario, both disturbances are Gaussian and independent:

$$\varepsilon_X \sim N(0, 1), \quad \varepsilon_Y \sim N(0, 1).$$

This case is included as the classical benchmark in which LiNGAM is not expected to identify the causal direction. In linear Gaussian models, the joint distribution is symmetric with respect to the two possible directions. Hence, even though the true data-generating process is $X \rightarrow Y$, the reverse representation $Y \rightarrow X$ can generate the same observational distribution.

Therefore, this scenario isolates the role of non-Gaussianity. If LiNGAM performs around random guessing, this confirms the standard non-identifiability of linear Gaussian SEMs.

3.2 Scenario B: Non-Gaussian independent disturbances

The next scenario was design to test whether identification is restored when the core assumptions of LiNGAM are satisfied. Disturbances are i.i.d. and non-Gaussian, satisfying all LiNGAM assumptions. We consider three distributions:

- **Uniform:** $\varepsilon \sim \mathcal{U}(-\sqrt{3}, \sqrt{3})$, unit variance, symmetric, light-tailed.
- **Laplace:** $\varepsilon \sim \text{Laplace}(0, 1/\sqrt{2})$, unit variance, symmetric, heavy-tailed.
- **Exponential:** $\varepsilon = (E - 1)/\sqrt{\text{Var}(E)}$, $E \sim \text{Exp}(1)$, unit variance, asymmetric.

Considering these three distributions allows us to distinguish different forms of non-Gaussianity and the role of tail behavior and skewness. If all three cases perform well, this suggests that no single moment such as skewness or kurtosis alone drives identification. Rather, what matters is that the distribution breaks the Gaussian symmetry.

3.3 Scenario C: Common volatility

Disturbances are generated as

$$\varepsilon_X = \sigma U_1, \quad \varepsilon_Y = \sigma U_2, \quad (3)$$

where U_1, U_2 are i.i.d. from one of the distributions in Scenario B (Gaussian, Uniform, Laplace, Exponential), and $\sigma = |\mathcal{N}(0, 1)| + 0.3 > 0$ is a shared random volatility factor independent of (U_1, U_2) . Conditional on σ , the errors are independent; marginally, they are positively dependent.

This scenario models economic settings where individual shocks are amplified by a common level of aggregate uncertainty. We also consider a *conditional* variant in which σ is observed, so the investigator can normalize:

$$\tilde{X} = X/\sigma, \quad \tilde{Y} = Y/\sigma, \quad (4)$$

restoring the standard LiNGAM structure $\tilde{X} = U_1, \tilde{Y} = \beta U_1 + U_2$. This experiment tests whether the failure in the common-volatility case is caused by the latent factor, or by the distributional shape of U_1, U_2 alone.

We vary the distribution of U_1 and U_2 :

- **Gaussian:** elliptical benchmark;
- **Uniform:** symmetric, light-tailed, non-Gaussian;
- **Laplace:** symmetric, heavy-tailed, non-Gaussian;
- **Exponential:** asymmetric, non-Gaussian.

This case is important because it separates two issues. On the one hand, the shocks are not independent marginally, which violates LiNGAM. On the other hand, the dependence is not arbitrary: it comes from a latent common scale factor. Finally, we also consider a robustness check in which U_1 and U_2 are allowed to follow different distributions. This introduces additional distributional asymmetry between the two variables. We treat this as a secondary check, since the main focus is on the case where U_1 and U_2 have the same distribution and dependence arises only through σ .

3.4 Scenario D: Mean independence but not independence

This experiment allows us to test whether LiNGAM can handle weaker forms of exogeneity that are common in econometric models. The final scenario studies a weaker condition than full independence. We define the structural error as

$$\varepsilon_Y = \gamma X Z,$$

where $Z \perp X$ and $\mathbb{E}[Z] = 0$. Then

$$\mathbb{E}[\varepsilon_Y \mid X] = \mathbb{E}[\gamma X Z \mid X] = \gamma X \mathbb{E}[Z] = 0.$$

Thus, the error is mean independent of X . However, it is not fully independent of X , because the full distribution of ε_Y still depends on X . For example, its conditional variance is

$$\text{Var}(\varepsilon_Y \mid X) = \gamma^2 X^2 \text{Var}(Z),$$

which varies with X .

We take X from a standardized Student- $t(5)$ distribution and vary:

- The distribution of Z : $\mathcal{N}(0, 1)$, Uniform, Laplace.
- The strength parameter $\gamma \in \{0.3, 1.0, 2.0\}$ (for Normal and Laplace Z), which controls the magnitude of the higher-order dependence. (Small values of γ correspond to weak dependence, while larger values correspond to stronger dependence.)

4 Results

4.1 Scenario A: Gaussian baseline

Table 1: DirectLiNGAM success rates: Gaussian baseline.

n	$\Pr(\hat{\pi} = X \rightarrow Y)$	$\Pr(\hat{\pi} = Y \rightarrow X)$
100	0.443	0.557
300	0.473	0.527
1000	0.530	0.470

Table 1 confirms the theoretical prediction. Success rates hover near 0.5 at all sample sizes and do not converge to any particular value, consistent with the known non-identifiability of linear Gaussian SEMs. The small fluctuations around 0.5 reflect finite-sample Monte Carlo variability rather than any systematic directional signal.

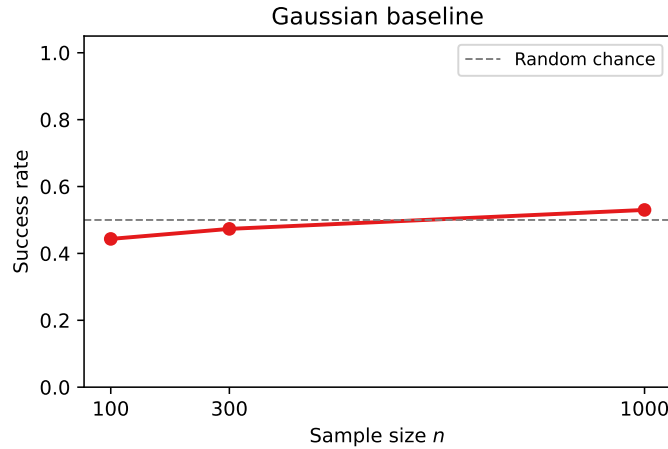


Figure 1: Success rate of DirectLiNGAM in the Gaussian baseline. The dashed line marks chance level (0.5). Success rates remain near 0.5 for all sample sizes, confirming non-identifiability.

4.2 Scenario B: Non-Gaussian independent disturbances

Table 2 and Figure 2 show that DirectLiNGAM achieves near-perfect identification across all three distributions when the independence assumption is satisfied. Performance is already excellent at $n = 100$ and exact at $n = 300$ or above for all distributions. Notably, the uniform distribution, which is symmetric and has negative excess kurtosis, performs as well as the asymmetric exponential.

Table 2: DirectLiNGAM success rates: non-Gaussian independent shocks.

Distribution	$n = 100$	$n = 300$	$n = 1000$
Uniform	1.000	1.000	1.000
Laplace	0.960	1.000	1.000
Exponential	1.000	1.000	1.000

This indicates that LiNGAM does not specifically require skewness: any non-Gaussian non-elliptical structure is sufficient for identification.

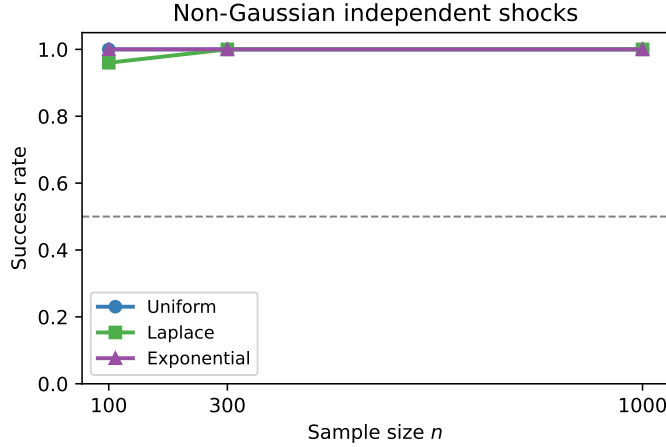


Figure 2: Success rates for non-Gaussian independent shocks. All three distributions converge rapidly to perfect identification.

4.3 Scenario C: Common volatility

Table 3: DirectLiNGAM success rates: common volatility (unobserved σ).

Distribution of U	$n = 100$	$n = 300$	$n = 1000$
Gaussian	0.540	0.493	0.473
Uniform	0.027	0.000	0.000
Laplace	0.930	0.990	1.000
Exponential	0.937	0.997	1.000

Table 3 and Figure 3 reveal a striking heterogeneity: the effect of common volatility depends fundamentally on the distribution of the idiosyncratic shocks U_1, U_2 .

Gaussian U . Success rates hover near 0.5, mirroring the Gaussian baseline. As noted in Section 2.2, the product σU_i with $U_i \sim \mathcal{N}(0, 1)$ produces an elliptical joint distribution. No non-Gaussian structure survives to enable identification.

Uniform U . Performance collapses dramatically: at $n = 300$ and $n = 1000$, DirectLiNGAM *always* selects the wrong direction (success rate = 0). This illustrates the theoretical failure described in Mesters et al. (2024): the common volatility factor introduces fourth-order dependence ($\text{Cov}(\varepsilon_X^2, \varepsilon_Y^2) > 0$) that flips the sign of the JADE objective, causing the algorithm to prefer the reversed DAG.

Laplace and Exponential U . Despite the same common volatility structure, DirectLiNGAM recovers the correct direction with high accuracy, converging to perfect identification as $n \rightarrow 1000$. The heavy tails of the Laplace distribution (excess kurtosis = 3) and the skewness of the exponential provide enough non-Gaussian asymmetry in the higher-order cumulants to overcome the distortion from the common factor.

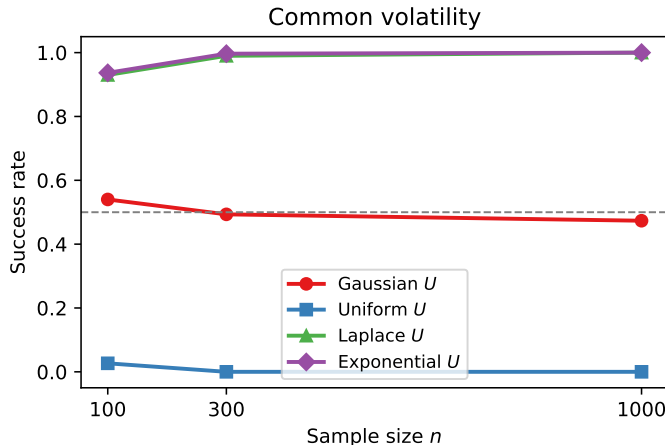


Figure 3: Success rates under common volatility for different distributions of the idiosyncratic shocks U . Gaussian U performs at chance; Uniform U exhibits systematic reversal; Laplace and Exponential U converge to perfect identification.

Effect of observing σ . Table 4 and Figure 4 show that conditioning on the volatility factor σ —i.e. normalizing by it when it is observed—restores perfect identification for both uniform and exponential idiosyncratic shocks.

Table 4: DirectLiNGAM success rates: common volatility with observed σ .

Distribution of U	$n = 100$	$n = 300$	$n = 1000$
Uniform	1.000	1.000	1.000
Exponential	1.000	1.000	1.000

This confirms that the failure in the latent- σ case is caused entirely by the unobserved common factor, not by any fundamental incompatibility between the distributional shape and the LiNGAM identification strategy.

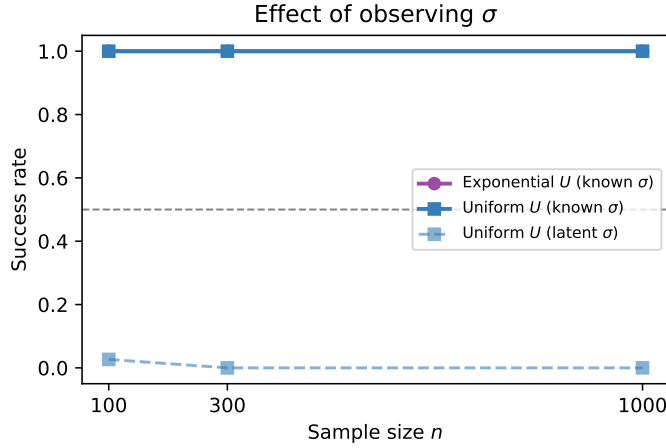


Figure 4: Comparison between latent and observed volatility σ . When σ is observed and the investigator normalizes by it, perfect identification is restored. Dashed line: latent Uniform U case for reference.

4.4 Scenario D: Mean independence

Table 5 and Figure 5 document the most severe failure mode. When the structural error $\varepsilon_Y = \gamma XZ$ satisfies mean independence but not full independence:

Table 5: DirectLiNGAM success rates: mean independence ($\gamma = 1$).

Distribution of Z	$n = 100$	$n = 300$	$n = 1000$
Normal	0.067	0.020	0.000
Uniform	0.000	0.000	0.000
Laplace	0.367	0.513	0.603

- For Normal and Uniform Z , DirectLiNGAM systematically selects the *wrong* direction $Y \rightarrow X$, and performance deteriorates rather than improves with sample size. At $n = 1000$, the success rate is 0.00 for Normal Z and 0.00 for Uniform Z . This is a *structural* failure: more data makes the algorithm more confidently wrong.
- For Laplace Z , performance is closer to chance and weakly increases with n , reaching 0.60 at $n = 1000$. The heavy tails of the Laplace distribution provide a partial residual non-Gaussian signal, but not sufficient for reliable identification.

4.4.1 Effect of dependence strength γ

Table 6 and Figure 6 report how performance varies with the dependence strength parameter $\gamma \in \{0.3, 1.0, 2.0\}$.

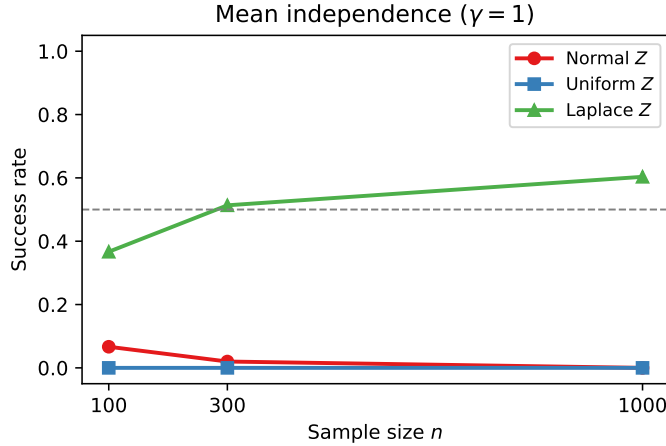


Figure 5: Success rates under mean independence ($\gamma = 1$). Normal and Uniform Z produce systematic reversal of the causal direction that worsens with sample size. Laplace Z shows partial improvement but remains unreliable.

Table 6: DirectLiNGAM success rates under mean independence: varying γ for Normal Z (left) and Laplace Z (right).

γ	Normal Z			Laplace Z		
	$n = 100$	$n = 300$	$n = 1000$	$n = 100$	$n = 300$	$n = 1000$
0.3	0.197	0.087	0.017	0.447	0.483	0.610
1.0	0.047	0.013	0.003	0.447	0.493	0.627
2.0	0.183	0.107	0.060	0.427	0.507	0.610

For Normal Z , the failure is present across all values of γ and does not diminish as $\gamma \rightarrow 0$. Even at $\gamma = 0.3$, the algorithm fails systematically at large samples. This rules out the interpretation that the failure is driven by the *strength* of the dependence; rather, it reflects a directional bias introduced by the departure from independence itself. For Laplace Z , success rates are stable across γ at roughly 0.44–0.63, with no clear monotone relationship to the dependence strength. These findings align with the theoretical argument in [Mesters et al. \(2024\)](#) that the failure of LiNGAM under mean independence is structural and cannot be alleviated by attenuating the violation.

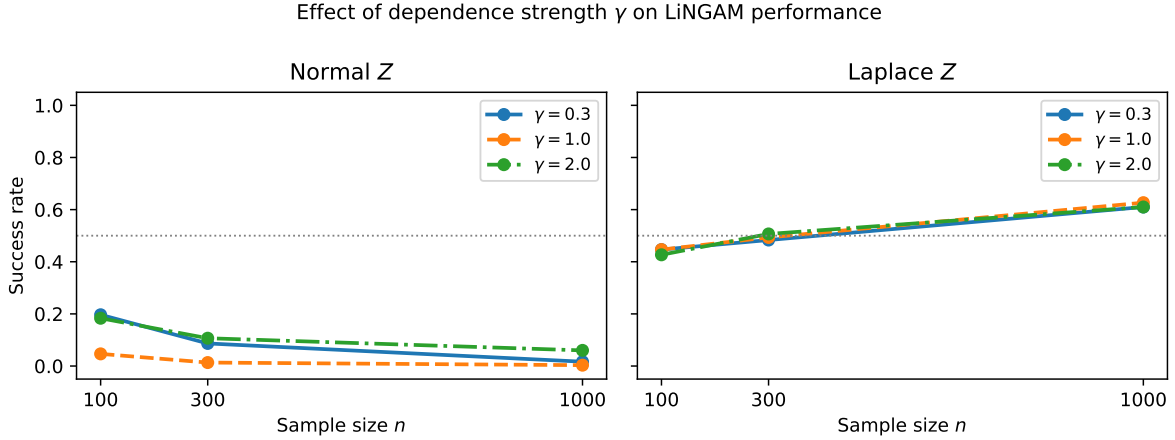


Figure 6: Effect of dependence strength γ on DirectLiNGAM under mean independence, for Normal Z (left) and Laplace Z (right). Failure is present across all γ values and does not improve with sample size.

5 Connection to Recent Theoretical Work

Recent theoretical work, particularly [Mesters et al. \(2024\)](#), shows that widely used causal discovery methods such as LiNGAM can systematically fail when the data exhibit features that are very common in macroeconomics and finance — namely, shared volatility across shocks. Intuitively, even if shocks are ‘unpredictable on mean’, they can still co-move in their volatility (for example, becoming more volatile under uncertainty). The literature shows that in these cases, standard algorithms may misinterpret co-movement as evidence of a different causal link, leading to biased or even reversed conclusions.

Our project is built upon the theoretical framework discussed above, and taking advantage of the simulation scenarios can perfectly isolate each of the main mechanisms that theory backed. At the same time we highlight the importance of emerging alternatives from the literature —such as mean-independence-based-approaches— that relax strong assumptions of the traditional methods and perform reliably in these settings. With this in mind [Mesters et al. \(2024\)](#) cleverly introduce LiMIAM model, which replaces mutual dependence with order-dependent mean independence $\mathbb{E}[\varepsilon_i | \varepsilon_j] = 0, \forall j < i$. Under this weaker condition, the causal graph is generically identifiable, and the DirectLiMIAM algorithm achieves correct causal discovery in simulations with dependent noise. Our results motivate the adoption of such methods in applied settings where common volatility or heteroskedasticity are likely.

6 Conclusion

We have studied the robustness of DirectLiNGAM through a comprehensive simulation study. Our findings yield three main conclusions:

Non-Gaussianity enables identification, provided errors are independent Across all non-Gaussian distributions considered — uniform, Laplace, and exponential — DirectLiNGAM achieves near-perfect causal recovery when disturbances are mutually independent, even at moderate sample sizes. Neither skewness nor heavy tails are necessary in isolation; any departure from

Gaussian symmetry structure is sufficient.

The structure of dependence, not its mere presence, determines whether identification fails. Under common volatility, whether LiNGAM succeeds depends on the distributional shape of the shocks. Elliptical(Gaussian) and light-tailed symmetric (Uniform) shocks lead to failure; heavy tailed and asymmetric shocks allow identification. Crucially the failure in the uniform case is caused entirely by the *unobserved* nature of the common scale factor: conditioning on the volatility restores perfect identification.

Mean independence is insufficient for independence-based methods When the structural error is mean-independent but not fully independent, DirectLiNGAM fails structurally. The algorithm not only fails to identify the causal link but reverses it in some scenarios with increasing confidence in the mistake as $n \rightarrow \infty$. this failure persists across distributional shapes and values of the dependence strength parameter. confirming that it reflects a fundamental limitation of the independence-based identification strategy.

These findings motivate the use of weaker identification conditions. The LiMIAM framework which relies on mean independence offers a theoretical grounded and empirically robust alternative for causal discovery in settings with dependent non-Gaussian disturbances.

References

- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.
- Cardoso, J.-F. and Souloumiac, A. (1996). Jacobi angles for simultaneous diagonalization. *SIAM Journal on Matrix Analysis and Applications*, 17(1):161–164.
- Comon, P. (1994). Independent component analysis, a new concept? *Signal Processing*, 36(3):287–314.
- Engle, R. F. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of UK inflation. *Econometrica*, 50(4):987–1007.
- Hyvärinen, A. (1999). Fast and robust fixed-point algorithms for independent component analysis. *IEEE Transactions on Neural Networks*, 10(3):626–634.
- Hyvärinen, A. and Smith, S. M. (2013). Pairwise likelihood ratios for estimation of non-Gaussian structural equation models. *Journal of Machine Learning Research*, 14:111–152.
- Ikeuchi, T., Ide, M., Zeng, Y., Maeda, T. N., and Shimizu, S. (2023). Python package for Causal Discovery based on LiNGAM. *Journal of Machine Learning Research*, 24(14):1–8.
- Lanne, M., Meitz, M., and Saikkonen, P. (2017). Identification of structural vector autoregressions via non-Gaussianity. *Journal of Business & Economic Statistics*, 35(3):466–477.
- Mesters, G., Ribot, A., Seigal, A., and Zwiernik, P. (2024). Causal discovery under mean independence and linearity. Working paper.
- Shimizu, S., Hoyer, P. O., Hyvärinen, A., and Kerminen, A. (2006). A linear non-Gaussian acyclic model for causal discovery. *Journal of Machine Learning Research*, 7:2003–2030.

Shimizu, S., Inazumi, T., Sogawa, Y., Hyvärinen, A., Kawahara, Y., Washio, T., Hoyer, P. O., and Bollen, K. (2011). DirectLiNGAM: A direct method for learning a linear non-Gaussian structural equation model. *Journal of Machine Learning Research*, 12(33):1225–1248.